SIH26162 INTERACTIVE GUIDE

Data Ingestion & Preprocessing Resource Guide

All links and API endpoints are directly clickable / touchable on mobile and desktop PDF readers.

1. Thermal Radiance & Hotspot Ingestion (NASA FIRMS)

NASA FIRMS API Key Registration & Documentation

- Portal: https://firms.modaps.eosdis.nasa.gov/api/
- Key Generator: https://firms.modaps.eosdis.nasa.gov/api/map_key

Get a free 32-character MAP_KEY to query real-time thermal anomalies and historical fire archives.

API Type / Endpoint	Instrument / Source	Format / Rate Limit	Primary Use Case
/api/area/ (Bounding Box)	VIIRS_SNPP_NRT, VIIRS_NOAA20_NRT, MODIS_NRT	CSV / GeoJSON / KML (10 req/min, 5000 pts/req)	Real-time bounding box ingestion around ROI / India boundary.
/api/country/ (Country Feed)	VIIRS_SNPP_NRT (375m resolution)	Direct ISO-3 Country Stream (IND for India)	National monitoring pipeline cron jobs (runs every 3-6 hours).
FIRMS Archive Download	VIIRS (VNP14IMGTDL) & MODIS (MCD14DL)	Shapefile, GeoPackage, Global Daily CSVs	Historical baseline modeling (3-5 years) for persistent flare clustering.

Key FIRMS Fields for Preprocessing

- `latitude`, `longitude`: Hotspot centroid | • `bright_ti4` (K): 375m I4 band brightness temp (VIIRS)
- `frp` (MW): Fire Radiative Power — primary metric for fire severity vs routine flare baseline
- `confidence`: Detection quality (low/nominal/high) | • `daynight`: 'D' vs 'N' (Night acquisitions have higher SNR for industrial flares)

2. Contextual & Infrastructure Data (OpenStreetMap & LULC)

OpenStreetMap Overpass API

- Live Endpoint: https://overpass-api.de/api/interpreter
- Interactive Query GUI: https://overpass-turbo.eu/

Query industrial land-use boundaries, refinery tags, power plants, flare stacks, and forest reserves.

Overpass QL: Industrial Sites

```
[out:json][timeout:30];
(
  way["landuse"="industrial"](bbox);
  relation["landuse"="industrial"](bbox);
  node["power"="plant"](bbox);
  way["power"="plant"](bbox);
  way["man_made"="works"](bbox);
  node["man_made"="flare"](bbox);
);
out body; >; out skel qt;
```

Overpass QL: Forest / Agriculture

```
[out:json][timeout:30];
(
  way["landuse"="farmland"](bbox);
  way["landuse"="farmyard"](bbox);
  way["landuse"="forest"](bbox);
  way["natural"="wood"](bbox);
  way["boundary"="national_park"](bbox);
);
out body; >; out skel qt;
```

Data Source	Direct Clickable Access URL	Preprocessing Use Case
Geofabrik OSM Extracts	download.geofabrik.de/asia/india.html	Offline bulk `.osm.pbf` ingestion into PostGIS with `osm2pgsql`.
ESA WorldCover (10m LULC)	esa-worldcover.org/en	10m global land cover raster for pixel-wise validation (Crop vs Built vs Forest).
Dynamic World (Google/WRI)	dynamicworld.app (Via Earth Engine API)	Near real-time 10m LULC probability distributions.

3. High-Resolution Multispectral Satellite Data (Optical & SWIR)

Microsoft Planetary Computer STAC API

- STAC Catalog: https://planetarycomputer.microsoft.com/api/stac/v1
- SDK Documentation: https://planetarycomputer.microsoft.com/docs/

Free access to Sentinel-2 L2A and Landsat 8/9 Cloud Optimized GeoTIFFs (COGs) with zero egress fees.

Copernicus Data Space Ecosystem (CDSE)

- Web Portal: https://dataspace.copernicus.eu/
- STAC / OData API: https://catalogue.dataspace.copernicus.eu/odata/v1/

Official ESA portal for full Sentinel-2 MSI data streams and Level-2A imagery.

Key Spectral Bands & Spectral Index Calculations

Spectral Index / Band	Band Formula / Wavelength	Detection & Classification Objective
SWIR-2 / SWIR-1	B12 (2190 nm) & B11 (1610 nm) [20m]	High-radiance thermal signatures, active combustion & industrial flare cores.
NBR (Normalized Burn Ratio)	`(B8_NIR - B12_SWIR2) / (B8_NIR + B12_SWIR2)`	Quantifies burn scars, structural devastation, and active fire perimeters.
NDVI (Vegetation Index)	`(B8_NIR - B4_Red) / (B8_NIR + B4_Red)`	Distinguishes agricultural crop residue burning from industrial structures.
BAIS2 (Burned Area Index)	`(1 - sqrt((B6*B7*B8A)/B4)) * ((B12-B8A)/(sqrt(B12+B8A)+1))`	Optimized burned area detection post-disaster validation.

4. Core Python Preprocessing Stack & Pipeline Architecture

Essential Geospatial Python Libraries

- GeoPandas & Shapely: Spatial joins (`sjoin`) between FIRMS points and OSM polygons.
- Rasterio & Xarray: Windowed reads of Sentinel-2 COG rasters for SWIR/RGB extraction.
- PySTAC Client: Querying satellite scenes by spatio-temporal bounds.
- DBSCAN / HDBSCAN: Spatio-temporal clustering to build the Persistent Source Heat Index.

Database & Backend Setup

- PostgreSQL 16 + PostGIS 3.4: Spatial indexing with `GIST` on geometries.
- osm2pgsql: High-speed OSM PBF parser to relational PostGIS tables.
- Redis + Celery: Background workers polling FIRMS NRT endpoints every 30 mins.
- FastAPI: Async REST endpoints for dashboard visual layers.